CHAPTER 3 · WEEK 2

Energy and Speed Are Related

How are energy and speed related?

Measurement / Data

REQUIRED

BQ

Big Question

How are energy and speed related?

Turn and talk. We will collect evidence before we answer.

REQUIRED

BQ

Big Question

My first thinking

Write one idea. There is no missing word yet.

REQUIRED

N1

One Ball, Two Speeds

One ball.

A gentle catch, then a hard catch.

Which catch did more to your hand?

REQUIRED

V1

Vocabulary — Speed

speed

a measurement of the _____ an object travels over an amount of time.

Reader p. 13

REQUIRED

V1

Vocabulary — Speed

Write the missing word on V1.

speed — a measurement of the _____ an object travels over an amount of time.

REQUIRED

V1

Vocabulary — Speed

speed — a measurement of the distance an object travels over an amount of time.

REQUIRED

N1

One Ball, Two Speeds

One ball at two speeds.

You decide from what you observe.

BALL → OBSERVE → EVIDENCE

REQUIRED

N1

One Ball, Two Speeds

Write both missing words.

Compare the _____ ball at two speeds.

A bigger _____ is how we know the faster ball had more energy of motion.

REQUIRED

N1

One Ball, Two Speeds

Compare the same ball at two speeds.

A bigger change is how we know the faster ball had more energy of motion.

REQUIRED

DEMO

Ramp and Target Preview

Preview. One ball.

Low setup, then a steeper setup.

Which cup move was bigger?

Do not measure. Do not write a speed number.

REQUIRED

N2

What We Measured

Tomorrow we measure the ramp height and how far the target moves.

We use centimeters.

We do not write a speed number.

REQUIRED

BQ

Big Question

Exit. One sentence about one ball.

Not graded.

REQUIRED

BQ

Big Question

How are energy and speed related?

What have we already observed?

REQUIRED

N2

What We Measured

A fair test changes one thing.

Keep one ball.

REQUIRED

N2

What We Measured

Write the missing words.

We measure ramp _____ in centimeters.

We measure how far the _____ moves in centimeters.

We do not measure a speed number.

The one thing we change is ramp _____.

We keep the _____ ball.

REQUIRED

N2

What We Measured

We measure ramp height in centimeters.

We measure how far the cup moves in centimeters.

We do not measure a speed number.

The one thing we change is ramp height.

We keep the same ball.

REQUIRED

LAB

Ramp It Up 2.0

Calibration trial.

Roll the ball once.

Do not record that roll.

REQUIRED

LAB

Ramp It Up 2.0

Measure the vertical ramp for 1 book, 2 books, and 3 books.

Write the centimeters.

REQUIRED

LAB

Ramp It Up 2.0

Three recorded trials at each setup.

Nine recorded trials.

Write how far the cup moves, in centimeters.

REQUIRED

LAB

Ramp It Up 2.0

Stop recording.

Averages and the graph come next.

REQUIRED

BQ

Big Question

How are energy and speed related?

REQUIRED

LAB

Ramp It Up 2.0

Average the three cup measurements at each setup.

REQUIRED

GRAPH

Ramp Height and Average Movement

Graph axes

Ramp height (cm)

Average cup movement (cm)

Do not label an axis as speed.

REQUIRED

GRAPH

Ramp Height and Average Movement

Plot your three averages.

Use your own measurements.

REQUIRED

LAB

Ramp It Up 2.0

Challenge

Move the cup as close as you can to 30 cm without going past.

REQUIRED

LAB

Ramp It Up 2.0

You may change the ramp.

Keep one ball.

Do not write a speed number.

REQUIRED

LAB

Ramp It Up 2.0

Which setup moved the cup farther?

A farther cup move is a bigger change.

REQUIRED

BQ

Big Question

Reader pages 15 and 16.

How are energy and speed related?

REQUIRED

N3

Steeper Ramp

A steeper ramp. One ball.

The Reader tells us what happens when the same ball rolls down a steeper ramp.

Watch how far the cup or bucket moves.

REQUIRED

N3

Steeper Ramp

Write the missing words.

Objects pick up more _____ on steeper ramps.

From the steeper ramp, that ball made a _____ change.

REQUIRED

N3

Steeper Ramp

Objects pick up more speed on steeper ramps.

From the steeper ramp, that ball made a greater change.

REQUIRED

N4

The Bounce

A soft bounce. A hard bounce.

Which bounce went farther up?

REQUIRED

N4

The Bounce

Write the missing word.

The faster ball bounced _____ than the slower ball.

REQUIRED

N4

The Bounce

The faster ball bounced higher than the slower ball.

A louder sound is also a bigger change.

REQUIRED

CHECK

Your Thinking

Check your thinking.

How are energy and speed related?

Name the ball you kept and one change that was bigger.

REQUIRED

CHECK

Your Thinking

Sample: That ball had more energy of motion when it caused a greater change.

I did not measure a speed number.

REQUIRED

BQ NOW

Big Question Now

Now I think

How are energy and speed related?

Write your answer.

REQUIRED

BQ NOW

Big Question Now

Exit. Not graded.

Look for one ball and a bigger change.

DROP IT

Drop It

FLEX · Drop It

Same ball. Two drop heights. Compare the craters.

This does not teach a new idea.

RECOVER

Friday Recovery

FLEX

Finish the notes, practice, or take the quiz.

The quiz is not a first lesson.